BioStack Protocol Packet

Provider-review draft for phased metabolic, repair, longevity, and cognitive observability

Version 2: adds Epitalon, total-materials planning, bacteriostatic-water math, and print page breaks

Note: Educational and observational draft only. This is not medical advice, not a prescription, and not injection instruction. All dosing, route, reconstitution, labs, imaging, and escalation decisions should be reviewed with a qualified provider. BioStack-style calculators should be treated as math checks only and verified manually.

1. Quick-Reference Phase Schedule

Weeks	Primary move	Why this timing makes sense
1-2	NAD+ only	Pure tolerance window. Keep the rest quiet so early energy, sleep, GI, mood, and injection-site signals are readable.
3	Add MOTS-c	Only add after NAD+ tolerance is clean. Treat as experimental mitochondrial/metabolic support, not guaranteed mitochondrial restoration.
4	Add GHK-Cu	Earlier add as a standalone repair/collagen/skin/connective-tissue signal. Do not add BPC/KPV the same week.
5-7	Add BPC-157 + KPV	Repair and anti-inflammatory layer after GHK-Cu has its own observation window. Avoid stacking all repair peptides on the same start date.
8-10	Consider retatrutide	Only after provider review, baseline labs, stable GI status, and quiet prior stack. Retatrutide becomes the dominant signal once introduced.
11-12	Consider Semax/Selank	Only if retatrutide is stable and sleep/mood/appetite are not actively shifting.
13-15+	Consider Epitalon, then Dihexa only if justified	Epitalon fits best as a later sleep/circadian/longevity-oriented short cycle. Dihexa stays last and highest caution.

2. Phase Explanations

Phase 1: NAD+ Foundation and Mitochondrial Build, Weeks 1-6

Goal: restore the metabolic and redox baseline first, then layer mitochondrial support after NAD+ tolerance is clear. This phase should feel intentionally boring at first. Boring is data.

- Weeks 1-2: NAD+ only. Do not add MOTS-c, repair peptides, cognitive peptides, or Epitalon yet.
- Week 3: add MOTS-c only if NAD+ tolerance is clearly good.
- Weeks 5-6: move toward the higher NAD+ maintenance range only if lower ranges are well tolerated.
- Track energy AM/PM, sleep quality, clarity/mood, digestion, injection-site response, training recovery, and resting heart rate if available.

Phase 2: Early GHK-Cu and Repair Layer, Weeks 4-7

Goal: introduce GHK-Cu earlier as its own clean intervention, then add BPC-157 and KPV after attribution is preserved. This avoids the “repair stack grenade launcher” problem.

- Week 4: introduce GHK-Cu as a standalone layer after NAD+ and MOTS-c are stable.
- Week 5 or 6: add BPC-157 and KPV after GHK-Cu has a quiet observation window.
- Rationale: GHK-Cu is less likely to dominate appetite, glucose, or GI signals than retatrutide, but it still deserves its own observation window.
- Avoid starting GHK-Cu, BPC-157, and KPV in the same week. Mechanistically attractive, observability nightmare.

Phase 3: Metabolic Optimization with Retatrutide, Weeks 8-13+

Goal: introduce retatrutide only after the metabolic and repair base is stable. Retatrutide is the most signal-dominant compound in the packet and should not be treated like just another small add-on.

- Do not start before provider review, recent labs, and stable GI status.
- Track appetite, hydration, constipation/diarrhea, nausea, reflux, abdominal pain, sleep, mood, resting heart rate, training performance, and weight trend.
- If liver or lipid markers worsen, treat that as a review trigger rather than pushing through.

Phase 4: Cognitive, Circadian, and Longevity Layer, Weeks 11-15+

Goal: add nootropic, sleep/circadian, or longevity-oriented peptides only after the metabolic layer is not actively moving the goalposts.

- Semax/Selank: consider only once retatrutide is stable and sleep/mood are clean.
- Epitalon: best placed after retatrutide has a stable low-dose rhythm and before any Dihexa decision. Use as a short, clearly bounded cycle rather than an open-ended add.
- Dihexa: highest caution. Keep later, short, and provider-supervised. Do not introduce while other variables are moving.

3. Decision Gates

Gate	Continue only if
Add MOTS-c	NAD+ has no persistent adverse signal.
Add GHK-Cu	NAD+ + MOTS-c are stable; no unresolved sleep, mood, HR, GI, or injection-site issue.
Add BPC/KPV	GHK-Cu has had a quiet observation window and there are no unresolved inflammatory, sleep, mood, or GI issues.
Add retatrutide	Recent labs reviewed, provider approval, no active GI issues, stable hydration/electrolytes, and resting HR acceptable.
Add Semax/Selank	Retatrutide is stable and not causing sleep, mood, anxiety, or appetite-overrestriction issues.
Add Epitalon	Semax/Selank are either not being used or are stable; sleep/circadian tracking is in place.
Consider Dihexa	All other variables are stable. Provider has reviewed risk/benefit. Use only if there is a specific reason, not just because it exists.

4. Total Materials Planning, Weeks 1-15

These totals are planning math based on the draft protocol through Week 15. They are not medical instructions. Maintenance beyond Week 15 should be calculated separately by multiplying the weekly amount by the number of additional weeks.

Bacteriostatic-water math: total bac water in mL = total mg needed / selected concentration in mg per mL. Actual vial-by-vial reconstitution depends on vial size, route, concentration comfort, and provider/pharmacy instructions.

Substance	Planning window	Draft usage assumption	Total mg needed	Bac water planning examples	Notes
NAD+	Weeks 1-15	W1-2: 25 mg 2x/wk; W3-4: 50 mg 2-3x/wk; W5-6: 75-100 mg 2-3x/wk; W7-15: 100 mg 2-3x/wk	2,400-3,700 mg	At 50 mg/mL: 48-74 mL total bac water. At 100 mg/mL: 24-37 mL.	Large-volume planning item. Confirm concentration, comfort, and route with provider.
MOTS-c	Weeks 3-15	Baseline planning: 5 mg 2x/wk. Higher optional path materially changes totals.	130 mg baseline; up to ~390 mg if 10 mg 3x/wk from W5-15	At 10 mg/mL: 13-39 mL.	Experimental; FDA has flagged limited human exposure and peptide-quality concerns.
GHK-Cu	Weeks 4-15	1 mg daily; possible titration to 1.5-2 mg daily after tolerance	84-168 mg	At 10 mg/mL: 8.4-16.8 mL. At 20 mg/mL: 4.2-8.4 mL.	Earlier add, but keep standalone for one week before BPC/KPV.
BPC-157	Weeks 5-15	250-500 mcg daily	19.25-38.5 mg	At 5 mg/mL: 3.85-7.7 mL. At 10 mg/mL: 1.93-3.85 mL.	Limited human evidence and sourcing/regulatory concerns.
KPV	Weeks 5-15	200-500 mcg daily	15.4-38.5 mg	At 5 mg/mL: 3.08-7.7 mL. At 10 mg/mL: 1.54-3.85 mL.	SubQ or oral route decision belongs with provider.
Retatrutide	Weeks 8-15 example only	Example conservative ramp: W8-9 0.5-1 mg/wk; W10-11 1-2 mg/wk; W12-13 2-4 mg/wk; W14-15 4-6 mg/wk	15-26 mg for this 8-week ramp	At 10 mg/mL: 1.5-2.6 mL. At 20 mg/mL: 0.75-1.3 mL.	Investigational. Use actual prescribed ramp for purchasing math.
Semax	Weeks 11-15	200-600 mcg/day	7-21 mg	At 5 mg/mL: 1.4-4.2 mL. At 10 mg/mL: 0.7-2.1 mL.	Often nasal in practice; route/form changes material planning.
Selank	Weeks 11-15	200-500 mcg/day	7-17.5 mg	At 5 mg/mL: 1.4-3.5 mL. At 10 mg/mL: 0.7-1.75 mL.	Do not start if sleep, anxiety, or mood are unstable.
Epitalon	Week 13+ short cycle	Provider-defined short cycle. Planning example: 1 mg/day for 10 days or 2 mg/day for 10 days.	10-20 mg per cycle	At 5 mg/mL: 2-4 mL. At 10 mg/mL: 1-2 mL.	Best placed after retatrutide stability and before any Dihexa decision. Evidence is mostly mechanistic/limited human.
Dihexa	Optional later only	No default included. Provider-defined short cautious cycle only.	TBD	TBD	Excluded from purchasing totals unless a provider-defined dose is selected.

5. Support Stack and Meal Anchors

Support item	Draft amount/timing	Purpose and caution notes
First meal	30-40 g protein + ~40 g clean carbs	Anchor first meal to reduce under-eating risk and support training/recovery.
L-carnitine	3 g/day	Supports fatty-acid transport into mitochondria. If TG/HDL worsens, provider may review increasing toward 4 g/day.
Phosphatidylcholine	3 g/day	Supports membrane phospholipid status and hepatic lipid handling.
Vitamin E	400 IU/day	Antioxidant support. Review bleeding risk, anticoagulants/antiplatelets, and total supplement load.
Selenium	200 mcg/day	Antioxidant/thyroid-related cofactor. NIH adult upper limit is 400 mcg/day from all sources.
TUDCA	500 mg with each of two biggest meals; 1,000 mg/day total	Bile-acid/liver support candidate during retatrutide. Frame as support, not guaranteed prevention of biliary sludge/cholestasis.
NAC	1,000 mg/day	Glutathione precursor; supports oxidative-stress handling.
Methyl B complex, B12, folate	Per label/provider	Methylation and liver-support context; avoid overcorrecting if labs are already high.
Magnesium glycinate	500 mg at night	Cofactor support; may help sleep. Adjust for GI tolerance and kidney status.
Zinc	30 mg/day	Immune and hepatic enzyme cofactor context. Consider copper balance if used long term.

Lipid Peroxidation, Plain-English Note

Lipid peroxidation is a free-radical chain reaction where reactive molecules steal electrons from lipids in cell membranes. That can damage membrane structure, change membrane fluidity, generate toxic byproducts, and contribute to inflammatory and ferroptosis-related pathways. The support-stack goal is not to pretend oxidation can be eliminated. The practical goal is to reduce unnecessary oxidative burden while labs and symptoms are monitored.

Alcohol Dehydrogenase Note

Alcohol dehydrogenase is an enzyme group found primarily in the liver and stomach lining. It catalyzes the conversion of ethanol into acetaldehyde, the first major step in alcohol metabolism. Zinc is one relevant cofactor, but alcohol avoidance or strict minimization is the cleaner move during a complex metabolic protocol.

6. Blood Work and Imaging Before Starting

Baseline testing gives the protocol a scoreboard. The goal is to detect metabolic dysfunction, hepatic stress, biliary risk, nutrient issues, and cardiovascular risk before adding signal-heavy compounds.

Baseline item	Baseline item	Baseline item
Fasting insulin	Fasting glucose	HOMA-IR
ALT	AST	ALP
GGT	Total bilirubin	Direct bilirubin
Albumin	Total protein	Triglycerides
HDL	LDL	Ferritin
Complete metabolic panel	Lipid panel	HbA1c
Vitamin D	B12	Folate
Selenium	Liver ultrasound with elastography for baseline hepatic steatosis and fibrosis	

Repeat Labs Every 5-6 Weeks During Retatrutide Escalation

Repeat lab	Repeat lab
Fasting insulin	GGT
ALT	AST
Triglycerides	HDL
Triglyceride/HDL ratio	

7. Provider-Review Adjustment Triggers

Signal	Review action
GGT rising or persistently elevated	Provider review. Consider whether retatrutide escalation is too aggressive for current hepatic/biliary processing capacity.
ALT/AST rising	Provider review for hepatic stress, training-related confounders, alcohol, medications, supplements, and retatrutide timing.
Triglyceride/HDL ratio worsening	Provider review of hepatic lipid export, protein intake, carbohydrate quality, L-carnitine plan, and retatrutide dose pace.
Persistent nausea, reflux, constipation, abdominal pain, or dehydration	Do not escalate retatrutide. Review dose, timing, hydration, fiber, electrolytes, gallbladder/pancreas warning signs, and provider guidance.
Resting HR trend rising after retatrutide	Hold escalation and review with provider. Phase 2 data reported dose-dependent HR increases.
Sleep, anxiety, mood, or cognition worsens	Do not add cognitive peptides or Epitalon until stable. Remove the most recent variable first if needed.

8. Daily and Weekly Tracking Templates

Date	Doses taken	Energy AM/PM	Sleep	Mood/focus	GI/appetite	HR/weight	Notes

Week	What changed?	Best signal	Worst signal	Labs/metrics	Decision for next week

9. Evidence and Safety Notes

- Retatrutide: investigational once-weekly triple hormone receptor agonist. Phase 2 obesity data reported dose-related gastrointestinal adverse events and dose-dependent heart-rate increases.
- Epitalon/Epitalon: pineal tetrapeptide discussed for telomerase, telomere, circadian, and aging biology. Evidence is not equivalent to a large modern FDA-approved clinical program; keep claims conservative and cycles bounded.
- BPC-157, MOTS-c, KPV, Semax, Selank, Epitalon, and related compounded peptides: regulatory and quality concerns include immunogenicity, impurity, API characterization, and limited human exposure information for some compounds.
- Vitamin E: NIH ODS describes antioxidant roles, but also notes bleeding-related cautions, medication interactions, and mixed trial outcomes at supplemental doses around 400 IU/day.
- Selenium: NIH ODS lists the adult tolerable upper intake level as 400 mcg/day. A 200 mcg supplement may be reasonable to discuss, but total intake matters.
- Ursodiol/TUDCA family: ursodeoxycholic acid is used for cholesterol gallstones and cholestatic liver diseases. TUDCA should be framed as bile-acid support, not guaranteed protection against biliary sludge or cholestasis.

Selected References

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